Asphericity excludes half-flat ALE bubbles

23 September 2026. This note strengthens the bounded-geometry result: for the actual aspherical shear manifolds, the assumed bound on the opposite Weyl curvature can be removed. Spin is not required. The key point is a compact-core embedding obstruction which treats both negative-definite ALE cores and the non-simply-connected rational-ball cores.

1. A topological obstruction to ALE cores

Lemma 1. Let M be a closed aspherical smooth oriented four-manifold. No sufficiently large compact core of a nonflat smooth Ricci-flat anti-self-dual ALE four-manifold X embeds smoothly in M.

The orientation on an ALE core is the one for which $W^+ = 0$. Reversing the orientation of an embedding does not change the non-embedding conclusion.

Proof. Take a large compact core $B \subset X$ with connected boundary $Y = S^3/\Gamma$ and cylindrical complement. Thus B has the homotopy type of X. Its fundamental group is finite. Since M is a closed aspherical manifold, $\pi_1(M)$ is torsion-free and its universal cover $\widetilde{M}$ is contractible. If $i : B \hookrightarrow M$ were an embedding, then $i_*\pi_1(B)$ would be trivial. The covering-space lifting criterion therefore lifts i to an embedding $\tilde{i} : B \hookrightarrow \widetilde{M}$. It is the core itself that lifts: no injectivity assertion about the fundamental group of a geometrical limit is being made. The lift is injective because its composition with the covering projection is the original embedding.

Wright's classification gives a negative-definite rational intersection form on $H_2(B;\mathbb{Q})$. If $b_2(B) > 0$, choose a nonzero rational class with negative square. Represent an integral multiple by a compact cycle and a transverse push-off in the interior of B. Their intersection number is unchanged by the embedding. In the contractible ambient manifold both classes are zero, so their intersection is zero. This is a contradiction.

It remains to treat $b_2(B) = 0$. Wright's theorem, combined with Suvaina's classification of Ricci-flat Kähler ALE surfaces, gives

$$\pi_1(B) \cong \mathbb{Z}_n, \qquad Y \cong L(n^2, nr-1), \qquad n \geqslant 2, \quad (n,r) = 1. \tag{1}$$

In particular, $H_1(B;\mathbb{Z}) = \mathbb{Z}_n$ and $H_1(Y;\mathbb{Z}) = \mathbb{Z}_{n^2}$. The nonflat assumption excludes the trivial Euclidean case $n = 1$.

Put $C = \widetilde{M} \setminus \operatorname{int} B$. Use small collar enlargements to apply ordinary Mayer–Vietoris to $\widetilde{M} = B \cup_Y C$; compactness of the ambient manifold is not required. Since its first and second homology vanish, the exact sequence gives an isomorphism

$$\mathbb{Z}_{n^2} = H_1(Y;\mathbb{Z}) \cong H_1(B;\mathbb{Z}) \oplus H_1(C;\mathbb{Z}) = \mathbb{Z}_n \oplus H_1(C;\mathbb{Z}). \tag{2}$$

The second factor must have order n. Therefore every element on the right is annihilated by n, while the left has an element of order n^2. This is impossible for every $n \geqslant 2$, composite or prime. The two cases exhaust the smooth nonflat ASD Ricci-flat ALE classification. This proves the lemma.

Why a spin-only argument would fail

There are spin rational-ball ALE spaces in (1). For example, take the three-centre Gibbons–Hawking space with centres in a regular triangle and its free cyclic quotient of order three supplied

by Wright's construction. The covering space is hyperkähler and spin. Pull-back on mod-two cohomology is injective for an odd-degree cover (transfer composed with pull-back is multiplication by three), so the quotient is also spin. It has $b_2 = 0$, fundamental group $\mathbb{Z}_3$ and a lens end of order nine. Lemma 1 excludes its embedding by (2), not by a false assertion that rational-ball bubbles are non-spin.

2. Half-Weyl bounds give full curvature control on aspherical types

Theorem 2. Fix $v > 0, D > 0, \Lambda > 0$ and $\chi_0 < \infty$. There is a constant $C(v, D, \Lambda, \chi_0)$ such that every closed aspherical oriented Einstein four-manifold satisfying

$$\mathrm{Ric} = -g, \quad V \geqslant v, \quad \mathrm{diam}(g) \leqslant D, \quad \|W^+\|_\infty \leqslant \Lambda, \quad \chi(M) \leqslant \chi_0 \tag{3}$$

has $\|\mathrm{Rm}\|_\infty \leqslant C$.

Proof. Suppose a sequence violates the conclusion and set $Q_j = \max|\mathrm{Rm}_{g_j}| \to \infty$, attained at p_j. Rescale by $\widehat{g}_j = Q_j g_j$. Then

$$|\mathrm{Rm}_{\widehat{g}_j}| \leqslant 1, \quad |\mathrm{Rm}_{\widehat{g}_j}|(p_j) = 1, \quad \mathrm{Ric}_{\widehat{g}_j} = -Q_j^{-1}\widehat{g}_j, \quad |W^+_{\widehat{g}_j}| \leqslant \Lambda/Q_j. \tag{4}$$

Relative Bishop comparison, using $V \geqslant v$ and diameter D, gives $\mathrm{Vol}_{g_j} B_r(x) \geqslant c(v, D)r^4$ for $0 < r \leqslant D$. After rescaling this holds on every fixed finite radius for all sufficiently large j. The curvature bound in (4) and this noncollapse give injectivity-radius control and smooth pointed Einstein compactness. The limit (X, g_∞, p) is complete, Ricci-flat, anti-self-dual, has Euclidean lower volume growth, and is nonflat because $|\mathrm{Rm}|(p) = 1$.

Einstein Chern–Gauss–Bonnet and $\chi \leqslant \chi_0$ give a uniform full L^2 curvature bound, unchanged by rescaling. The limit therefore has finite curvature energy. The classical Bando–Kasue–Nakajima asymptotic theorem makes it an ALE space. Equivalently this is the smooth deepest-bubble case of Einstein compactness. Choosing a global curvature maximum ensured a smooth limit; an unproved classification of intermediate orbifold bubbles is not needed.

Smooth pointed convergence embeds every fixed compact subset of X into M_j for sufficiently large j. Choose a large ALE core. Lemma 1 forbids precisely this embedding. This contradiction proves the theorem. No spin, simply-connected-bubble or global complex-gluing assumption has entered the proof.

3. A diameter-only spectral gap on the actual shear family

Let $M = M(q, g; u, f)$ be the self-gluing of $F_{q,2} \times \Sigma_g$, with $q \geqslant 1$, $g \geqslant 2$, under

$$(\phi, x) \longmapsto (-\phi + u(x), f(x)), \qquad f \in \mathrm{Diff}^+(\Sigma_g), \qquad 0 \neq [u] \in H^1(\Sigma_g; \mathbb{Z}).$$

These are the explicit self-gluings covered by the complete virtual Kähler and actual monopole arguments of Rounds 11–12; no arbitrary multi-block assembly is included by implication. No order condition on the mapping class of f is imposed. The inherited results give:

- M is aspherical and $\tau(M) = 0$;
- no finite cover is Kähler;
- $\pi_1(M)$ contains $\mathbb{Z}^2$;
- $\chi(M) = 4q(g-1)$, and an actual monopole class has square $A = 2\chi(M) > 0$.

As before, write

$$t = -3\lambda_1(W^+)/4, \quad b = \lambda_3(W^+) - \lambda_2(W^+), \quad \mathscr{S}(g) = \fint b^2.$$

Theorem 3. For every $D > 0$ and $0 \leqslant \theta < 1/\sqrt{3}$ there is $\varepsilon(D,\theta) > 0$ such that every normalized negative Einstein metric on any member of this actual shear family with

$$b \leqslant 4\theta t, \qquad \operatorname{diam}(g) \leqslant D \tag{5}$$

satisfies $\mathscr{S}(g) \geqslant \varepsilon(D,\theta)$. Zeros of W^+ are allowed. There is no lower amplitude, opposite Weyl, full-curvature, injectivity-radius or separately assumed noncollapse bound.

Proof. The previously established maximum principle, applied at a positive maximum of t, gives $t \leqslant [2(1-3\theta^2)]^{-1}$ and hence a uniform bound on W^+. At such a maximum the lowest eigenvalue is simple since $\lambda_2 - \lambda_1 \geqslant 2(1-\theta)t > 0$. The case $t \equiv 0$ needs no maximum argument.

The actual monopole scalar inequality gives

$$V \geqslant 4\pi^2\chi(M). \tag{6}$$

In this family $\chi(M) \geqslant 4$, so $V \geqslant 16\pi^2$. Bishop comparison gives a diameter-dependent upper volume bound. Consequently (6) also bounds $\chi(M)$ in terms of D. All constants in Theorem 2 are thus controlled by D, θ, uniformly across the family. Full curvature is bounded, and the geometric compactness used in the preceding note follows.

If splitting energies tended to zero, the smooth limit would be exact negative Einstein with at most two self-dual Weyl eigenvalues. Derdziński's classification gives a KE metric on a cover of degree at most two, or an ASD metric. The first contradicts the inherited virtual Kähler obstruction. The second, together with the preserved signature zero, has $W^- = 0$, so is real hyperbolic; Preissmann's theorem excludes its $\mathbb{Z}^2$ subgroup. This proves the uniform gap.

Corollary 4. Every exact Einstein sequence on these manifolds with fixed $\theta < 1/\sqrt{3}$ and $\mathscr{S}(g_j) \to 0$ satisfies

$$\operatorname{diam}(g_j) \to \infty.$$

This replaces the former alternative "diameter escape or opposite-Weyl blow-up" by diameter escape alone, in this actual aspherical family. It still permits arbitrary exact Einstein metrics outside the relative pinching class, or with a splitting bounded away from zero, and it does not establish their nonexistence.

4. Sources and precise status

1. Wright, *Quotients of gravitational instantons*, Theorem A, Remark 3 and Corollary C: https://arxiv.org/pdf/1102.2442.
2. Suvaina, *ALE Ricci-flat Kähler metrics and deformations of quotient surface singularities*, Theorem B, Theorem 2.2, Proposition 3.3 and its classification proof: https://arxiv.org/pdf/1102.2672. These give the cyclic fundamental group and the lens boundary in (1), not merely the existence of one chosen rational ball.
3. Bando–Kasue–Nakajima, *On a construction of coordinates at infinity on manifolds with fast curvature decay and maximal volume growth*, Invent. Math. 97 (1989), 313–349: https://doi.org/10.1007/BF01389045.

4. Derdziński, Proposition 5 and Theorem 2: https://www.numdam.org/article/CM_1983___49_3_405_0.pdf.
5. Actual objects: `round11_graph_topology.md`, `round11_kahler.md`, `round12_global.md`; spectral calculation and initial escape proof: `line1_einstein_spectral_application.md`, `line1_allowed_branch_escape.md`.

The classification and compactness inputs are classical. The compact-core exclusion is proved explicitly above, including the rational-ball case that a sphere-only or spin-only argument misses. No exhaustive publication-priority assertion is made. The result removes an actual curvature assumption from the preceding all-output spectral application; it does not claim to solve the unrestricted negative Einstein problem.